

Advanced (Habanero)

- Q1.** What is the effect of the transformation $y = f(x) + 2$ on the graph of $y = f(x)$?
- (A) Horizontal shift 2 units left
(B) Vertical shift 2 units up
(C) Horizontal shift 2 units right
(D) Vertical shift 2 units down
(E) Reflection over x-axis
- Q2.** Which transformation reflects the graph of $y = f(x)$ over the y-axis?
- (A) $y = f(-x)$
(B) $y = f(x) + 2$
(C) $y = -f(x)$
(D) $y = f(x) - 2$
(E) $y = f(x + 2)$
- Q3.** What is the effect of the transformation $y = 2f(x)$ on the graph of $y = f(x)$?
- (A) Vertical compression by a factor of 2
(B) Vertical stretch by a factor of 2
(C) Horizontal stretch by a factor of 2
(D) Horizontal compression by a factor of 2
(E) Reflection over x-axis
- Q4.** Which of the following is the graph of $y = f(x - 3)$?
- (A) Graph of $y = f(x)$ shifted 3 units left
(B) Graph of $y = f(x)$ shifted 3 units right
(C) Graph of $y = f(x)$ shifted 3 units up
(D) Graph of $y = f(x)$ shifted 3 units down
(E) Graph of $y = f(x)$ reflected over y-axis
- Q5.** What is the effect of the transformation $y = -f(x)$ on the graph of $y = f(x)$?
- (A) Reflection over x-axis
(B) Reflection over y-axis
(C) Vertical shift 2 units up
(D) Vertical shift 2 units down
(E) Horizontal shift 2 units left
- Q6.** Which transformation represents a horizontal compression by a factor of 2?
- (A) $y = f(2x)$
(B) $y = f(x/2)$
(C) $y = 2f(x)$
(D) $y = f(x) + 2$
(E) $y = f(x) - 2$
- Q7.** What is the effect of the transformation $y = f(x) - 1$ on the graph of $y = f(x)$?
- (A) Vertical shift 1 unit up
(B) Vertical shift 1 unit down
(C) Horizontal shift 1 unit left
(D) Horizontal shift 1 unit right
(E) Reflection over x-axis
- Q8.** Which of the following graphs represents the transformation $y = f(-x)$?
- (A) Graph of $y = f(x)$ reflected over y-axis
(B) Graph of $y = f(x)$ reflected over x-axis
(C) Graph of $y = f(x)$ shifted 2 units up
(D) Graph of $y = f(x)$ shifted 2 units down
(E) Graph of $y = f(x)$ shifted 2 units left

Open Ended Questions (FRQ)

- Q9.** Describe the transformations applied to the graph of $y = f(x)$ to obtain the graph of $y = 2f(x - 1) + 3$. Support your answer with $x \cdot y$ coordinate points.
- Q10.** Graph the function $y = f(x)$ and its transformation $y = -f(-x)$. Describe the effect of the transformation on the graph of $y = f(x)$. Use $x \cdot y$ coordinate points to support your answer.

Chef's Tips

- Tip 1: Remind students to identify the parent function.
- Tip 2: Emphasize the importance of describing the transformations in the correct order.
- Tip 3: Encourage students to use coordinate points to support their answers.